Random Walks

Experiment :- On $\mathbb{Z}$. Toss a fair coin (prob. of H. is $1/2$). If H turns up, move one step to right otherwise move step to the left. Keep on doing it

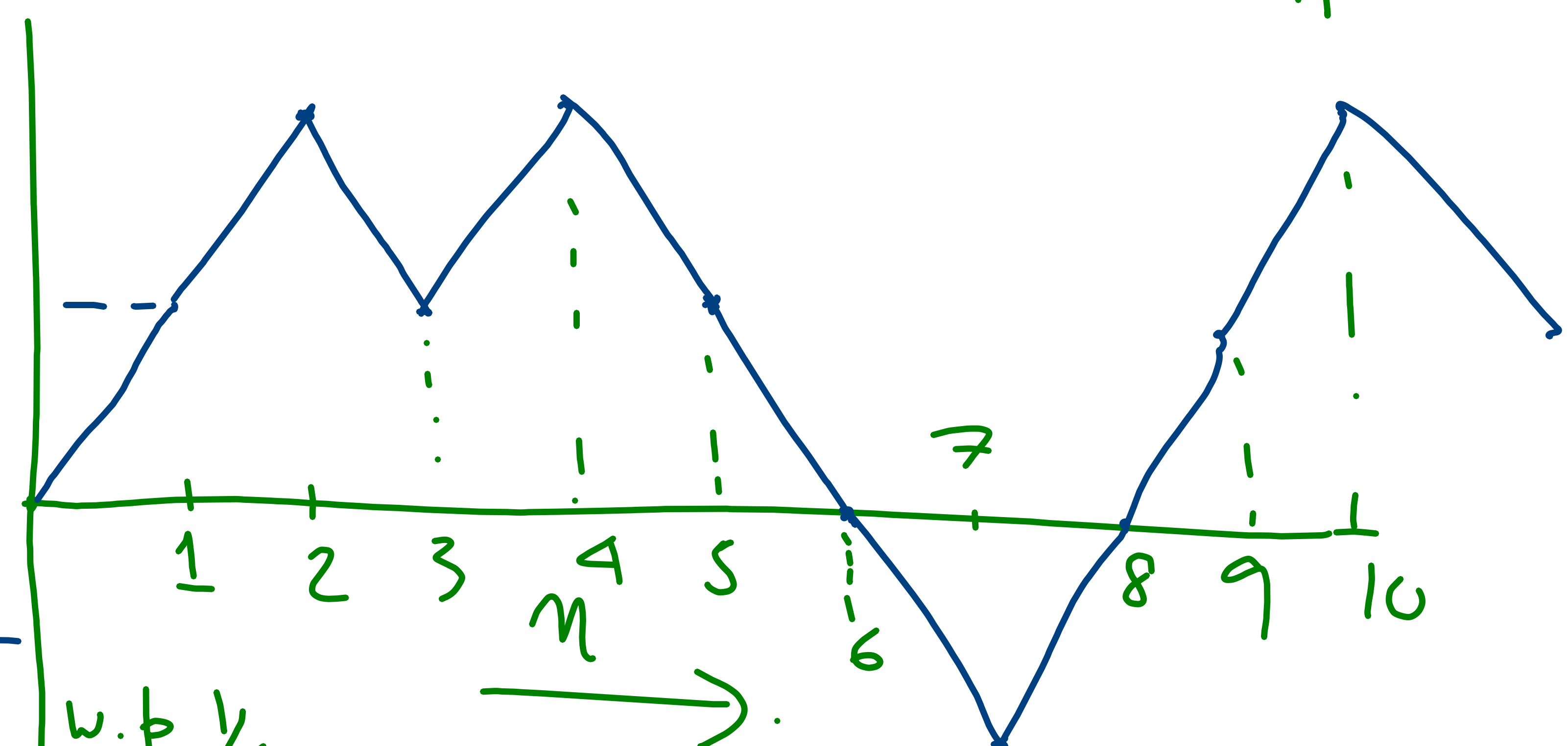
Define $\epsilon_i = \begin{cases} +1 & \text{w.p } 1/2 \\ -1 & \text{w.p. } 1/2 \end{cases}$.

represents the random step at the ith toss.

$\mathbb{Z}, E$

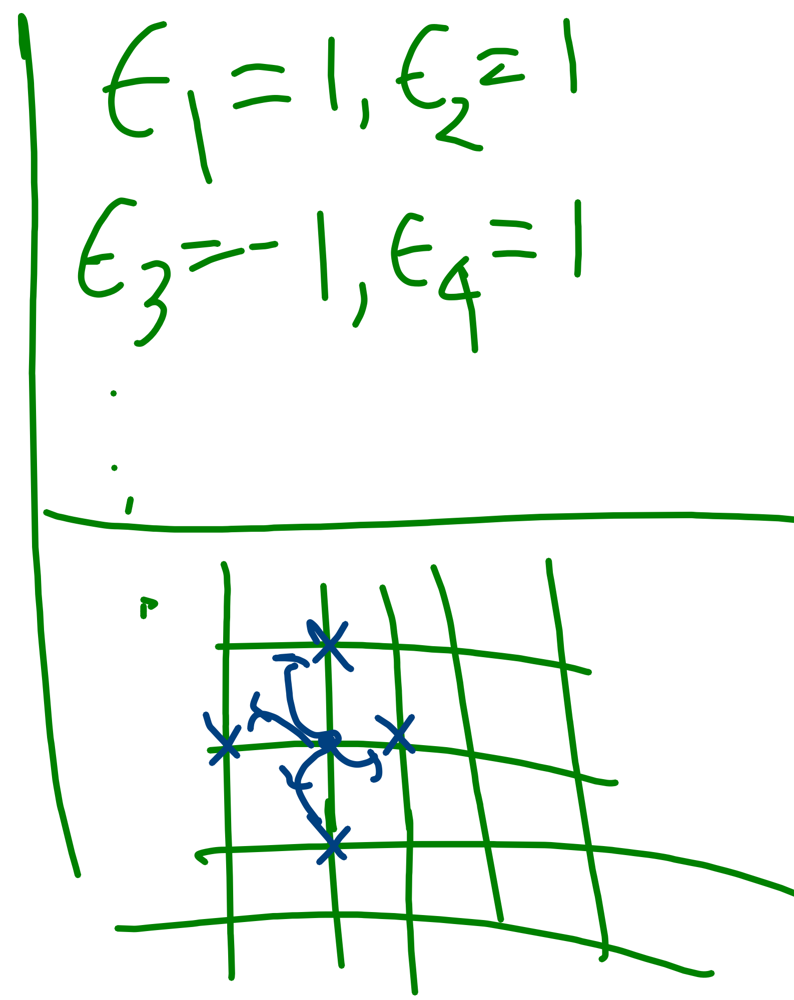
$S_0 = a \in \mathbb{Z}$

$S_n = a + \sum_{i=1}^{n} \epsilon_i$, position of the walker at time n.



$\epsilon_1 = 1, \epsilon_2 = 1$

$\epsilon_3 = -1, \epsilon_4 = 1$



$\epsilon_i = \begin{cases} +e_i & \text{w.p } 1/4 \\ -e_i & \text{w.p } 1/4 \end{cases}$

$\epsilon_1 = (1,0), \epsilon_2 = (0,1)$

$\{B_n : n \geq 1\}$ s.t. $B_n \cap B_m = \phi$, then $P(\bigcup_1^\infty B_n) = \sum_{n=1}^{\infty} P(B_n)$

Q. For a simple sym. r.w., starting at the origin, does it come back to the origin at some time with prob 1.

$S_0 = 0$. $A = \{S_n = 0$ for some $n \geq 1\}$

Want to show $P(A) = 1$.

Def. $A_n = \{S_n = 0\}$. $A = \bigcup_{n=1}^{\infty} A_n$.

Note, $A_n = \phi$ if n is odd.

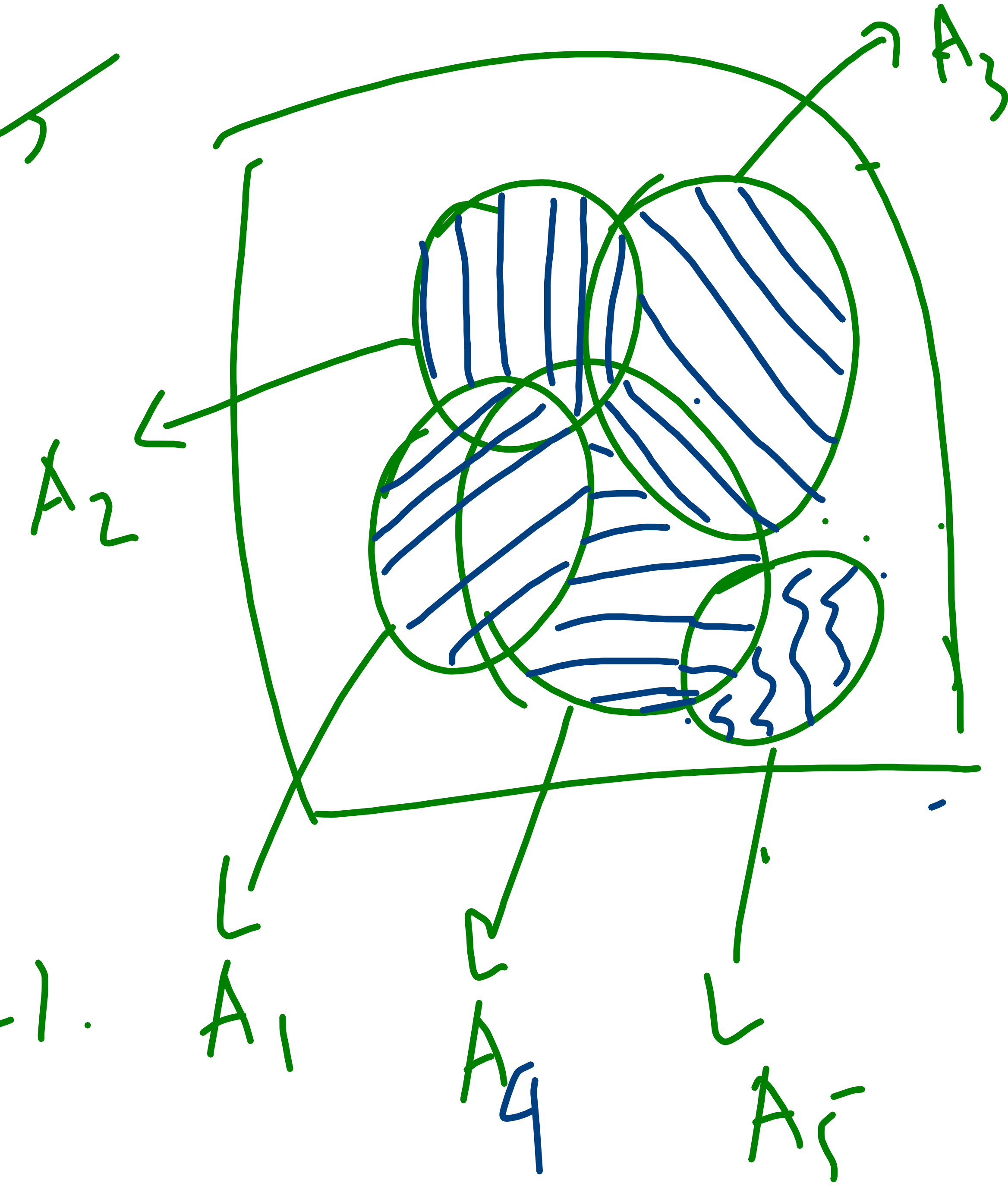
Define $B_1 = A_1$, for $n \geq 2$,

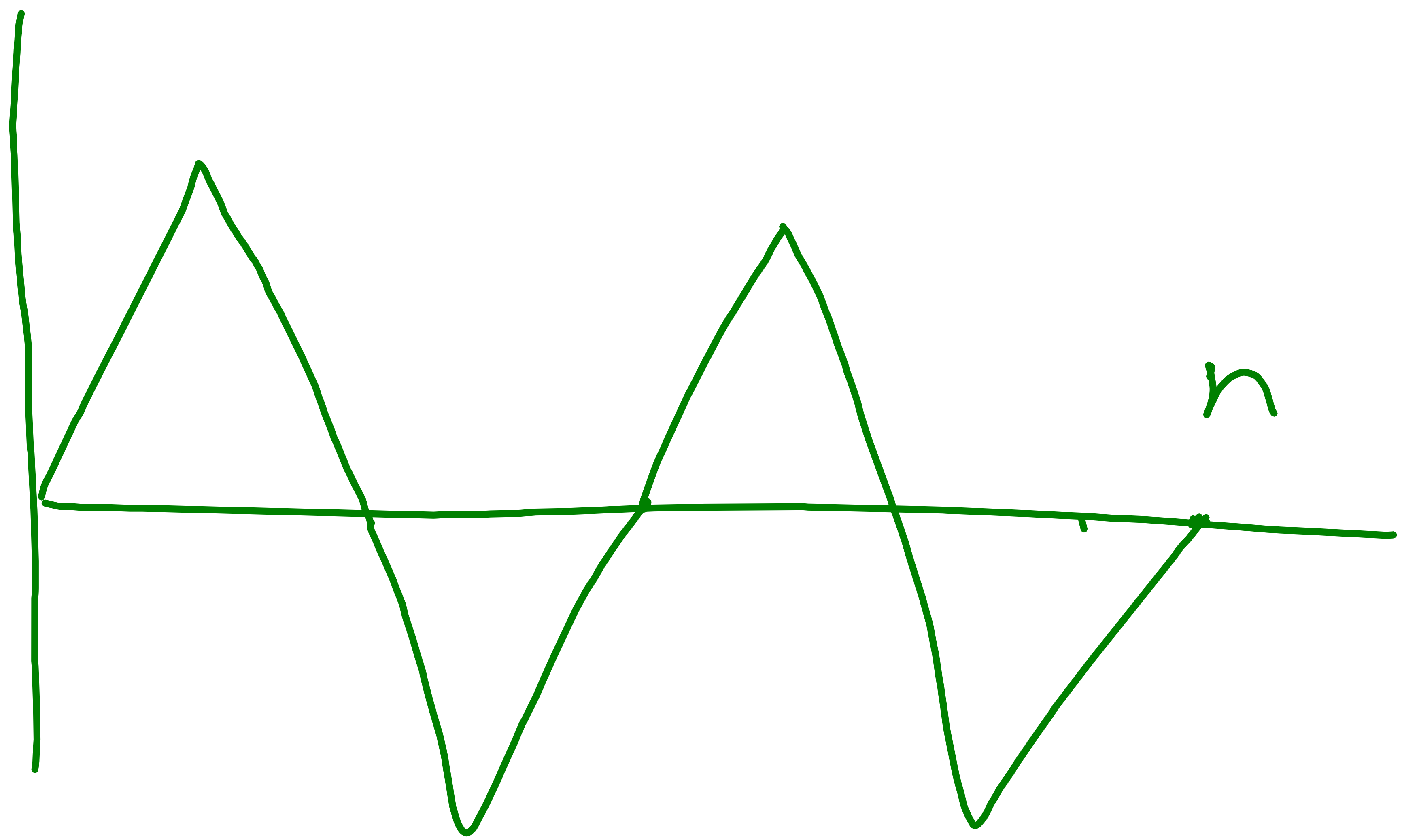
$$B_n = A_n \setminus \left(\bigcup_{k=1}^{n-1} A_n \right)$$

Note, $B_n \cap B_m = \phi$.

$$\bigcup_{k=1}^{n} A_k = \bigcup_{k=1}^{n} B_k \quad \forall\, n \geq 1.$$

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} B_k.$$





$$B_n = \{ S_n = 0,\ S_k \neq 0 \text{ for } k = 1, \dots, n-1 \}$$

$= \{$ R.W. hit 0 for the first time at $n \}$.

Notation :

$$P(A) = P\left(\bigcup_{n=1}^{\infty} A_n\right)$$
$$= P\left(\bigcup_{n=1}^{\infty} B_n\right)$$
$$= \sum_{n=1}^{\infty} P(B_n) = \sum_{n\geq 1}^{\infty} f_n = \sum_{n\geq 1}^{\infty} f_{2n}$$

$$u_n = P(S_n = 0)$$
$$f_n = P(S_n = 0,\ S_k \neq 0 \text{ for } k = 1, \ldots, n-1).$$

$u_n = f_n = 0$ for n odd

Find the gen fn $\{f_n\}$

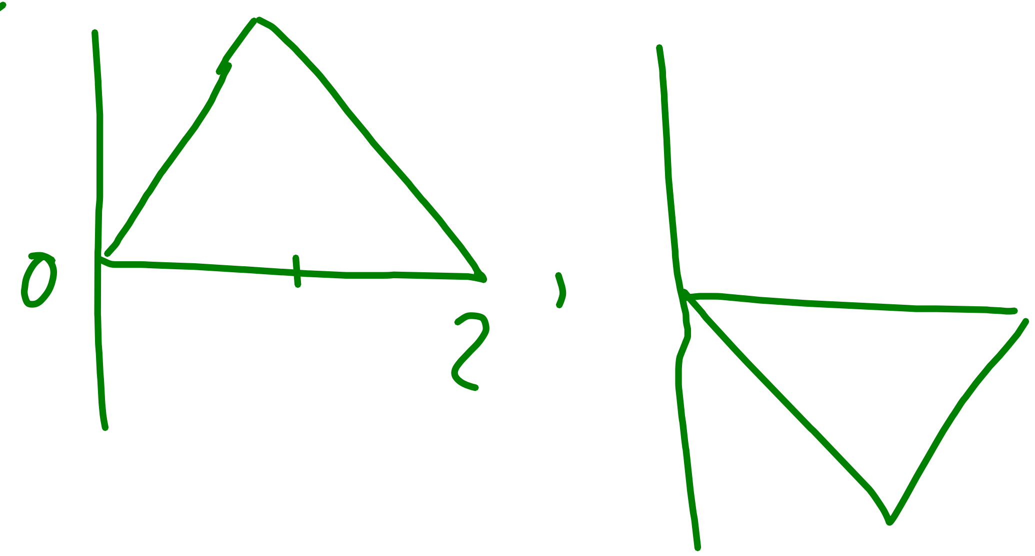
Prop^n: For $n \geq 1$.

$$f_{2n} = u_{2n-2} - u_{2n}.$$

$u_0 = P(S_0 = 0) = 1$.

Notation: $f_0 = 0$.

Calculation for $n=1$.

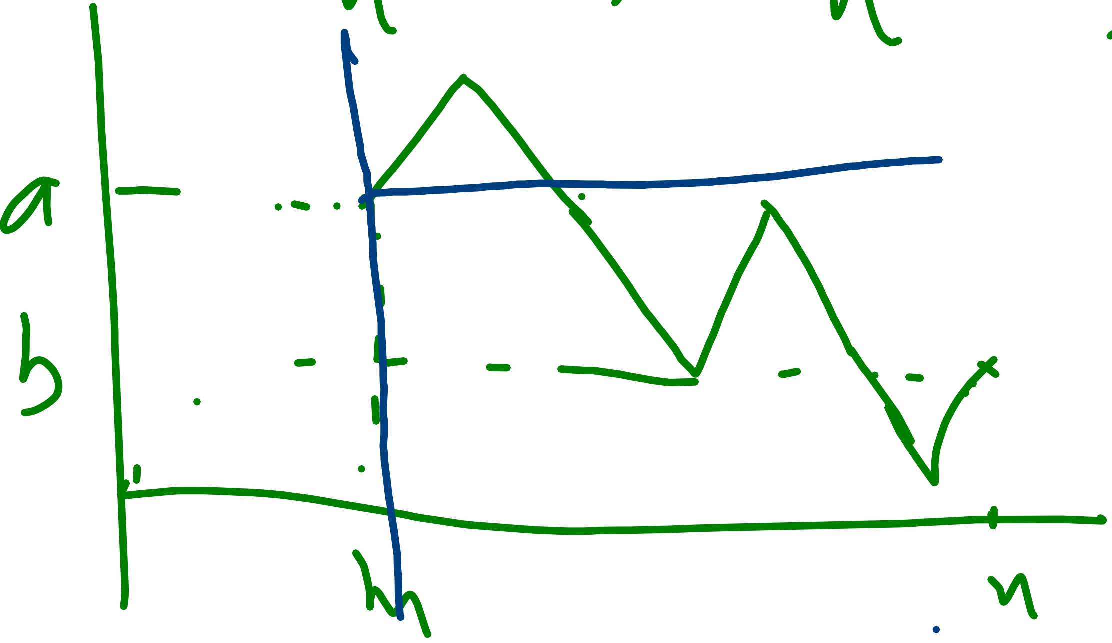
$f_2 = P\Big($  $\Big) = \frac{1}{2}$.

$= u_2$

$f_2 = \frac{1}{2} = 1 - \frac{1}{2}$

$= u_0 - u_2$

Def :- A path start from (m, a) and ending at (n, b) is a finite seq^n $s_m, s_{m+1}, \ldots, s_n$ s.t.
$s_m = a$, $s_n = b$, $|s_k - s_{k-1}| = 1$ for $k = m+1, \sim n$.



Not. : $N^{(m,a)}_{(n,b)}$ is the set of all paths st. from (m,a) and ending at (n,b). When $m=a=0$, we drop the superscript. Note,

$$N^{(m,a)}_{(n,b)} = N_{(n-m,\, b-a)}$$

$p \to$ No. of steps + 1

$q \to$ No. of step

$$|N_{n,a}| = \begin{cases} 0 & \text{if } n+a \text{ is odd} \\ \binom{n}{\frac{n+a}{2}} & \text{o.w.} \end{cases}$$

$$\left| \begin{array}{l} p+q = n \\ p-q = a \\ p = \frac{n+a}{2} \\ q = \frac{n-a}{2} \end{array} \right.$$

Any event B which involves $\epsilon_1, \dots, \epsilon_n$, has the prob.

$$P(B) = \frac{1}{2^n} \#\left(\text{No. of path statisfying the cond. } \cancel{\text{of}}\ B\right)$$

$$u_{2n} = \frac{1}{2^{2n}} |N_{2n,0}|$$

Find the gen. fn. of $\{u_n\}$

$$= \frac{1}{2^{2n}} \binom{2n}{n}$$

$$u_{2n}\sqrt{\pi n} = \frac{1}{2^{2n}} \cdot \frac{(2n)!}{n!\,n!} \sqrt{\pi n}$$

$$= \frac{(2n)!}{\sqrt{2\pi 2n}\,(2n)^{2n} e^{-2n}} \times \left(\frac{\sqrt{2\pi n}\, n^n e^{-n}}{n!}\right)^2 \times \frac{1}{\cancel{2^{2n}}} \cdot \cancel{\sqrt{\pi n}} \times \frac{\cancel{2^{2n}}}{\cancel{\sqrt{\pi n}}}$$

$\longrightarrow 1$ as $n \to \infty$.

Stirling's approximation

$$n! \sim \sqrt{2\pi n}\, n^n e^{-n}$$

We say $a_n \sim b_n$ if $\frac{a_n}{b_n} \to 1$ as $n \to \infty$

- $u_{2n}\sqrt{\pi n} \to 1$ as $n \to \infty$.
- $u_{2n} \sim \dfrac{1}{\sqrt{\pi n}}$.
- $u_{2n} \to 0$ as $n \to \infty$.
- $\sum_{n=0}^{\infty} u_{2n} = \infty$

Ex: $f_{2n} \sim \dfrac{1}{\sqrt{4\pi n^3}}$

$$P(A) = \sum_{n=1}^{\infty} f_{2n}.$$

$$\sum_{n=1}^{N} f_{2n} = \sum_{n=1}^{N} u_{2n-2} - u_{2n}$$

$$= u_0 - u_{2N}$$

$$= 1 - u_{2N}$$

$$\sum_{n=1}^{\infty} f_{2n} = \lim_{N\to\infty} (1 - u_{2N})$$

$$= 1$$

Define :-

T = time of return.

$= \inf\{n \geq 1 : S_n = 0\}$.

$E(T^{1/2-\epsilon}) < \infty$

$\forall \epsilon > 0$

$P(T = n) = P(S_n = 0, S_k \neq 0 \text{ for } k = 1, \dots n-1)$

$= f_n$.

$T < \infty$ a.s.

$E(T) = \sum_{n=0}^{\infty} P(T > n) = \sum_{n=0}^{\infty} u_{2n} = \infty$

$P(T > n)$

$= P\{\text{Returning after time } n\}$.

$= P\left(\bigcup_{k=n+1}^{\infty} B_k\right)$

$= \sum_{k=n+1}^{\infty} P(B_k)$

$= \sum_{k=n+1}^{\infty} f_k$.

$$\sum_{k=2n}^{\infty} f_k = f_{2n} + f_{2n+2} + \cdots$$

$$= \lim_{N\to\infty} \sum_{k=n}^{n+N} f_{2k}.$$

$$= \lim_{N\to\infty} \sum_{k=n}^{n+N} (u_{2k-2} - u_{2k})$$

$$= u_{2n-2} - \lim_{N\to\infty} \underbrace{u_{2n+2N}}_{\geq 0}$$

$$= u_{2n-2}.$$